TF-IDF and Cosine Similarity

Lab 5

Term Frequency (TF)

- **<u>TF: Term Frequency</u>**, which measures how frequently a term occurs in a document.
  - Since every document is different in length, it is possible that a term would appear much more times in long documents than shorter ones. Thus, the term frequency is often divided by the document length (aka. the total number of terms in the document) as a way of normalization.

    Normalized TF(t) = (Number of times term t appears in a document) / (Total number of terms in the document).
  - Measures **proportion of the document**

Example:

- Document A: 10 occurrences in 100 words → 0.10
- Document B: 10 occurrences in 1000 words → 0.01
  → Clearly more important in Document A.

Max-Normalized Term Frequency (TF)

$$\text{Max-Normalized TF} = \frac{TF(t,d)}{\max TF \text{ in } d}$$

- TF(t,d): frequency of your term
- maxTF(t′,d): frequency of the **most common term in the same document.**

- Example: A = 3 B= 2 C = 1
  - Maximum TF is 3
  - Normalized TF (B) = 2/3

- **Use Normalized TF when:**
  - Comparing **different document lengths**
  - You care about **true proportion**

- **Use Max-Normalized TF when:**
  - You care about **relative importance inside a document**
  - You want to reduce impact of **very frequent terms dominating**

Inverse Document Frequency (IDf)

- **<u>IDF: Inverse Document Frequency</u>**, which measures how important a term is **across a collection of documents**
  - While computing TF, all terms are considered equally important.
  - However it is known that certain terms, such as "is", "of", and "that", may appear a lot of times but have little importance.
  - Thus we need to weigh down the frequent terms while scale up the rare ones, by computing the following:

    IDF(t) = log(Total number of documents / Number of documents with term t in it).

  - IDF gives **higher weight to rare words** (more important) and **lower weight to common words (not important)**

To summarize

- **TF:** Measures how many times a **word appears** in the document.
- **IDF:** Represents how **common** the word is across the different documents.

tf-idf weight

- **<u>The tf-idf weight is composed of two terms:</u>** the first computes the normalized Term Frequency (TF), the second term is the Inverse Document Frequency (IDF).

$$TF\text{-}IDF = TF * IDF$$

- A word is important if:
  - It appears frequently in a document (high TF)
  - But rarely across documents (high IDF)

TF → "Is this word important in this document?"

IDF → "Is this word special across all documents?"

TF-IDF : Is this word both frequent here AND rare elsewhere?

tf-idf weight (Example)

- Consider a document containing **100 words** wherein the word ***cat*** appears **3 times**.
- The **term frequency** (i.e., tf) for ***cat*** is then **(3 / 100) = 0.03**.
- Now, assume we have **10 million documents** and the word ***cat*** appears in **one thousand** of these.
- Then, the **inverse document frequency** (i.e., idf) is calculated as **log(10,000,000 / 1,000) = 4**.
- Thus, the **Tf-idf weight** is the **product** of these quantities: **0.03 * 4 = 0.12**.

Summarize: TF-IDF gives high scores to words that are frequent in a document but rare across documents (more important)

Example 1

- we have 3 documents
- only 2 contain a certain keyword.
- the keyword appears 4 times in a 100 words document.
- Calculate its tf-idf weight?
- **<u>Solution</u>**
  - Tf=4/100
  - Idf= log(3/2)
  - Therefore, tf-idf= 4/100 * log(3/2) =0.04*0.176=0.07

TF-IDF Example 2

- Let's assume that we have a collection of **four** documents as follows:
- *d*1: "*The sky is blue.*
- *d*2: "*The sun is bright today.*"
- *d*3: "*The sun in the sky is bright.*"
- *d*4: "*We can see the shining sun, the bright sun.*"
- **Task:** Determine the tf-idf scores for each term in each document.

- **Step1:** Filter out the stopwords. After removing the stopwords, we have
  - *d*1: "*sky blue*
  - *d*2: "*sun bright today*"
  - *d*3: "*sun sky bright*"
  - *d*4: "*can see shining sun bright sun*"

- **Step2:** Compute TF, therefore, we find document-word matrix and then normalize the rows to sum to 1.

$$f_{t,d}$$

	blue	bright	can	see	shining	sky	sun	today
1	1	0	0	0	0	1	0	0
2	0	1	0	0	0	0	1	1
3	0	1	0	0	0	1	1	0
4	0	1	1	1	1	0	2	0

$$\mathtt{tf}(t,d) = \frac{f_{t,d}}{\sum_{t'} f_{t',d}}$$

	blue	bright	can	see	shining	sky	sun	today
1	1/2	0	0	0	0	1/2	0	0
2	0	1/3	0	0	0	0	1/3	1/3
3	0	1/3	0	0	0	1/3	1/3	0
4	0	1/6	1/6	1/6	1/6	0	1/3	0

- **Step3:** Compute IDF: Find the number of documents in which each word occurs, then compute the formula:

$$f_{t,d}$$

	blue	bright	can	see	shining	sky	sun	today
1	1	0	0	0	0	1	0	0
2	0	1	0	0	0	0	1	1
3	0	1	0	0	0	1	1	0
4	0	1	1	1	1	0	2	0
n_t	**1**	**3**	**1**	**1**	**1**	**2**	**3**	**1**

$$\mathtt{idf}(t, D) = \log_{10} \frac{N}{n_t}$$

$N = 4$

	blue	bright	can	see	shining	sky	sun	today
	0.602	0.125	0.602	0.602	0.602	0.301	0.125	0.602

$$\log_{10} \frac{4}{1} = 0.602 \qquad \log_{10} \frac{4}{3} = 0.125$$

Step4: Compute TF-IDF: Multiply TF and IDF scores.

$\mathtt{tf}(t, d)$

	blue	bright	can	see	shining	sky	sun	today
1	1/2	0	0	0	0	1/2	0	0
2	0	1/3	0	0	0	0	1/3	1/3
3	0	1/3	0	0	0	1/3	1/3	0
4	0	1/6	1/6	1/6	1/6	0	1/3	0

- TF-IDF: Multiply TF and IDF scores, use to rank importance of words within documents
  - Most important word for each document is highlighted

x

$\mathtt{idf}(t, D)$

	blue	bright	can	see	shining	sky	sun	today
	0.602	0.125	0.602	0.602	0.602	0.301	0.125	0.602

$\mathtt{tfidf}(t, d, D) = \mathtt{tf}(t, d) \cdot \mathtt{idf}(t, D)$

	blue	bright	can	see	shining	sky	sun	today
1	**0.301**	0	0	0	0	0.151	0	0
2	0	0.0417	0	0	0	0	0.0417	**0.201**
3	0	0.0417	0	0	0	**0.100**	0.0417	0
4	0	0.0209	**0.100**	**0.100**	**0.100**	0	0.0417	0

Example 3

- Through a step-by-step process, calculate TFIDF for the given corpus and mention the word(s) having highest value.

- Document 1: We are going to Mumbai
- Document 2: Mumbai is a famous place.
- Document 3: We are going to a famous place.
- Document 4: I am famous in Mumbai.

Solution

- **Term Frequency:** Term frequency is the frequency of a word in one document.

	We	Are	Going	to	Mumbai	is	a	famous	Place	I	am	in
D1	1	1	1	1	1	0	0	0	0	0	0	0
D2	0	0	0	0	1	1	1	1	1	0	0	0
D3	1	1	1	1	0	0	1	1	1	0	0	0
D4	0	0	0	0	1	0	0	1	0	1	1	1

- Inverse Document Frequency: The other half of TFIDF which is Inverse Document Frequency. For this, let us first understand what does document frequency mean. **Document Frequency** is the number of documents in which the word occurs **irrespective of how many times it has occurred in those documents.**

We	Are	going	to	Mumbai	is	a	Famous	place	I	am	in
2	2	2	2	3	1	2	3	2	1	1	1

- Talking about inverse document frequency, we need to put the document frequency in the denominator while the total documents is the numerator. Here, the total number of are 3, hence **inverse document frequency** becomes:

We	Are	going	to	Mumbai	is	a	Famous	place	I	am	in
4/2	4/2	4/2	4/2	4/3	4/1	4/2	4/3	4/2	4/1	4/1	4/1

- The formula of TFIDF for any word W becomes:
- TFIDF(W) = TF(W) * log (IDF(W))
- The words having highest value are – **Mumbai**, **Famous**.

Example 4

- Given a document with the terms A, B and C with the following frequencies A: 3, B: 2, C: 1 .
- The document belongs to a collection of 10,000 docs.
- The document frequencies are: A: 50, B:1300, C:250.
- Compute the normalized tf and the tf-idf and compare them. You could also check the effects of using normalized tf measures.
- The idf are as below:
  - A idf = log(10000/50) = 5.3;
  - B idf = log(10000/1300) = 2.0;
  - C idf = log(10000/250) = 3.7

Solution

- **A** tf = 3/3;
- idf = log(10000/50) = 5.3;
- tf-idf=5.3
- **B** tf = 2/3;
- idf = log(10000/1300) = 2.0;
- tf-idf=1.3
- **C** tf = 1/3;
- idf = log(10000/250) = 3.7;
- tf-idf=1.2

Example 5

- Here we are considering two documents, each containing the below sentences.
- **Document 1:** I read the SVM algorithm article in dataaspirant blog
- **Document 2:** I read the randomforest algorithm article in dataaspirant blog

- Calculate tf-idf weights?

Solution

- After the **text preprocessing** step, we will end up with the below sentences.
- **Document** 1: read SVM algorithm article dataaspirant blog
- **Document** 2: read randomforest algorithm article dataaspirant blog

TF-IDF Calculation Example							
Words	Count		Term Frequency (TF)		Inverse Document Frequency (IDF)	TF * IDF	
	Document 1	Document 2	Document 1	Document 2		Document 1	Document 2
read	1	1	0.17	0.17	0	0	0
svm	1	0	0.17	0	0.3	**0.05**	0
algorithm	1	1	0.17	0.17	0	0	0
article	1	1	0.17	0.17	0	0	0
dataaspirant	1	1	0.17	0.17	0	0	0
blog	1	1	0.17	0.17	0	0	0
randomforest	0	1	0	0.17	0.3	0	**0.05**

Results

- For the first document, it is about the **SVM algorithm**, whereas for the second document, it is about **randomforest algorithm**.
- This is clearly coming up in the **TF-IDF** calculation. For all other words, we are getting **0**, and for **SVM and randomforest** words, we are getting some value.

For students to solve

Exercise 1 Given the tables below

tf)

	Doc1	Doc2	Doc3
car	27	4	24
auto	3	33	0
insurance	3	33	0
best	14	0	17

idf)

term	df_t	idf_t
car	18,165	1.65
auto	6723	2.08
insurance	19,241	1.62
best	25,235	1.5

Compute the tf-idf weights for the terms in the tables for each document.

Tf normalization Take the values in the tf table above and replace them with normalized tf weights. Compute the tf-idf again. and compare the results.

Cosine similarity

- **Cosine Similarity** is a way to measure **how similar two documents (or vectors) are**, based on the **angle between them**, not their length.
- We define cosine similarity mathematically as the dot product of the vectors divided by their magnitude. For example, if we have two vectors, A and B, the similarity between them is calculated as:

$$similarity(A, B) = cos(\theta) = \frac{A \cdot B}{\|A\|\|B\|}$$

where

- θ is the angle between the vectors,
- $A \cdot B$ is dot product between A and B and calculated as $A \cdot B = A^T B = \sum_{i=1}^{n} A_i B_i = A_1 B_1 + A_2 B_2 + \ldots + A_n B_n$,
- $\|A\|$ represents the L2 norm or magnitude of the vector which is calculated as $\|A\| = \sqrt{A_1^2 + A_1^2 \ldots A_1^n}$.

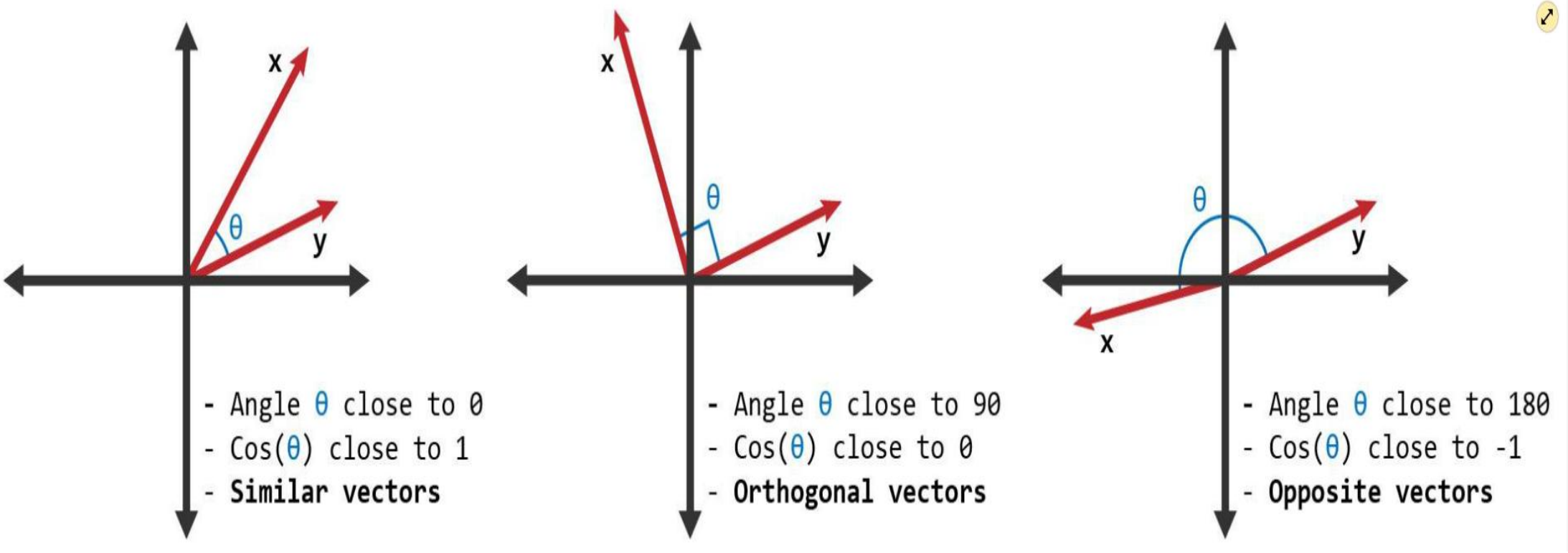
Cosine similarity (cont.)

- The similarity can take values between -1 and +1. Smaller angles between vectors produce larger cosine values, indicating greater cosine similarity.
- For example:
  - When two vectors have the same orientation, the angle between them is 0, and the cosine similarity is 1.
  - Perpendicular vectors have a 90-degree angle between them and a cosine similarity of 0.
  - Opposite vectors have an angle of 180 degrees between them and a cosine similarity of -1.

Smaller angle → more similar
Larger angle → less similar

Cosine similarity (cont.)

Smaller the angle, higher the similarity.



Example 1

- Calculate the cosine similarity of the two documents given below.
- Document 1 = 'the best data science course'
- Document 2 = 'data science is popular'

Solution

After creating a word table from the documents, the documents can be represented by the following vectors:

	the	best	data	science	course	is	popular
D1	1	1	1	1	1	0	0
D2	0	0	1	1	0	1	1

- $D1 = [1, 1, 1, 1, 1, 0, 0]$
- $D2 = [0, 0, 1, 1, 0, 1, 1]$

Using these two vectors we can calculate cosine similarity. First, we calculate the dot product of the vectors:

$$D1 \cdot D2 = 1 \times 0 + 1 \times 0 + 1 \times 1 + 1 \times 1 + 1 \times 0 + 0 \times 1 + 0 \times 1 = 2$$

Second, we calculate the magnitude of the vectors:

$$||D1|| = \sqrt{1^2 + 1^2 + 1^2 + 1^2 + 1^2 + 0^2 + 0^2} = \sqrt{5}$$

$$||D2|| = \sqrt{0^2 + 0^2 + 1^2 + 1^2 + 0^2 + 1^2 + 1^2} = \sqrt{4}$$

Finally, cosine similarity can be calculated by dividing the dot product by the magnitude

$$similarity(D1, D2) = \frac{D1 \cdot D2}{\|D1\|\|D2\|} = \frac{2}{\sqrt{5}\sqrt{4}} = \frac{2}{\sqrt{20}} = 0.44721$$

The angle between the vectors is calculated as:

$$cos(\theta) = 0.44721$$

$$\theta = \arccos(0.44721) = 63.435$$

Example 2

Table 2.5. Document Vector or Term-Frequency Vector

Document	*team*	*coach*	*hockey*	*baseball*	*soccer*	*penalty*	*score*	*win*	*loss*	*season*
Document1	5	0	3	0	2	0	0	2	0	0
Document2	3	0	2	0	1	1	0	1	0	1
Document3	0	7	0	2	1	0	0	3	0	0
Document4	0	1	0	0	1	2	2	0	3	0

<u>Calculate the cosine similarity between the first 2 documents in the above table?</u>

- **<u>Solution</u>**
- Cosine similarity between two term-frequency vectors
- Suppose that ***x*** and ***y*** are the first two term-frequency vectors in the previous table.
- That is, x=(5,0,3,0,2,0,0,2,0,0) and y=(3,0,2,0,1,1,0,1,0,1). To compute the cosine similarity between the two vectors, we get:

$$
\begin{aligned}
\boldsymbol{x}^t \cdot \boldsymbol{y} &= 5 \times 3 + 0 \times 0 + 3 \times 2 + 0 \times 0 + 2 \times 1 + 0 \times 1 + 0 \times 0 + 2 \times 1 \\
&\quad + 0 \times 0 + 0 \times 1 = 25 \\
||\boldsymbol{x}|| &= \sqrt{5^2 + 0^2 + 3^2 + 0^2 + 2^2 + 0^2 + 0^2 + 2^2 + 0^2 + 0^2} = 6.48 \\
||\boldsymbol{y}|| &= \sqrt{3^2 + 0^2 + 2^2 + 0^2 + 1^2 + 1^2 + 0^2 + 1^2 + 0^2 + 1^2} = 4.12 \\
sim(\boldsymbol{x}, \boldsymbol{y}) &= 0.94
\end{aligned}
$$

Thank You!